

linear search

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import time

# Function for Linear Search
def linear_search(arr, key):
    for i in range(len(arr)):
        if arr[i] == key:
            return i
    return -1

# User Input
n = int(input("Enter the number of elements: "))

arr = []
print("Enter the elements:")
for i in range(n):
    arr.append(int(input()))

key = int(input("Enter the element to search: "))

# Start Timer
start_time = time.perf_counter()

# Perform Linear Search
result = linear_search(arr, key)

# End Timer
end_time = time.perf_counter()

# Calculate Execution Time
execution_time = end_time - start_time

# Display Results
if result != -1:
    print(f"\nElement found at index {result}")
else:
    print("\nElement not found.")

print(f"\nExecution Time: {execution_time:.8f} seconds")

print("\nTime Complexity:")
print("Best Case    : O(1)")
print("Average Case  : O(n)")
print("Worst Case    : O(n)")

print("\nSpace Complexity: O(1)")

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Enter the number of elements: 5
Enter the elements:
34
67
09
87
5
Enter the element to search: 5

Element found at index 4

Execution Time: 0.00014242 seconds

Time Complexity:
Best Case    : O(1)
Average Case : O(n)
Worst Case   : O(n)

Space Complexity: O(1)

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Binary search

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import time

# Function for Binary Search
def binary_search(arr, key):
    low = 0

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high = len(arr) - 1

while low <= high:
    mid = (low + high) // 2

    if arr[mid] == key:
        return mid
    elif arr[mid] < key:
        low = mid + 1
    else:
        high = mid - 1

return -1

# User Input
n = int(input("Enter the number of elements: "))

arr = []
print("Enter the elements (in sorted order):")
for i in range(n):
    arr.append(int(input()))

key = int(input("Enter the element to search: "))

# Start Timer
start_time = time.perf_counter()

# Perform Binary Search
result = binary_search(arr, key)

# End Timer
end_time = time.perf_counter()

# Calculate Execution Time
execution_time = end_time - start_time

# Display Results
if result != -1:
    print(f"\nElement found at index: {result}")
else:
    print("\nElement not found.")

print(f"\nExecution Time: {execution_time:.8f} seconds")

print("\nTime Complexity:")
print("Best Case    : O(1)")
print("Average Case : O(log n)")
print("Worst Case   : O(log n)")

print("\nSpace Complexity: O(1)")

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Enter the number of elements: 5
Enter the elements (in sorted order):
45
7
86
0
4
Enter the element to search: 86

Element found at index: 2

Execution Time: 0.00011217 seconds

Time Complexity:
Best Case    : O(1)
Average Case : O(log n)
Worst Case   : O(log n)

Space Complexity: O(1)

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